

Geometric Action of a Quantum Channel on the Bloch Sphere

1 Introduction

In quantum information theory, the state of a single qubit can be represented geometrically using the *Bloch sphere*. Quantum channels, which describe physical processes such as noise, decoherence, or measurement, act on these states and transform them in characteristic geometric ways.

Figure 1 illustrates how a quantum channel maps pure states on the surface of the Bloch sphere to mixed states inside a deformed Bloch ball.

2 The Bloch Sphere Representation

Any single-qubit density matrix ρ can be written as

$$\rho = \frac{1}{2} (I + \vec{r} \cdot \vec{\sigma}), \quad (1)$$

where:

- $\vec{r} = (r_x, r_y, r_z) \in \mathbb{R}^3$ is the Bloch vector,
- $\vec{\sigma} = (\sigma_x, \sigma_y, \sigma_z)$ are the Pauli matrices,
- $\|\vec{r}\| \leq 1$.
- If $\|\vec{r}\| = 1$, the state is **pure** and lies on the surface of the Bloch sphere.
- If $\|\vec{r}\| < 1$, the state is **mixed** and lies inside the sphere.

3 Quantum Channels

A quantum channel $\mathcal{N}$ is a completely positive, trace-preserving (CPTP) map:

$$\mathcal{N} : \rho \rightarrow \mathcal{N}(\rho). \quad (2)$$

Physically, quantum channels describe:

- Noise (depolarization, dephasing)
- Energy loss (amplitude damping)
- Interaction with an environment

Mathematically, any qubit channel can be written as an affine transformation on the Bloch vector:

$$\vec{r} \longrightarrow \vec{r}' = A\vec{r} + \vec{t}, \quad (3)$$

where:

- A is a real 3×3 contraction matrix,
- $\vec{t}$ is a translation vector.

4 Geometric Interpretation

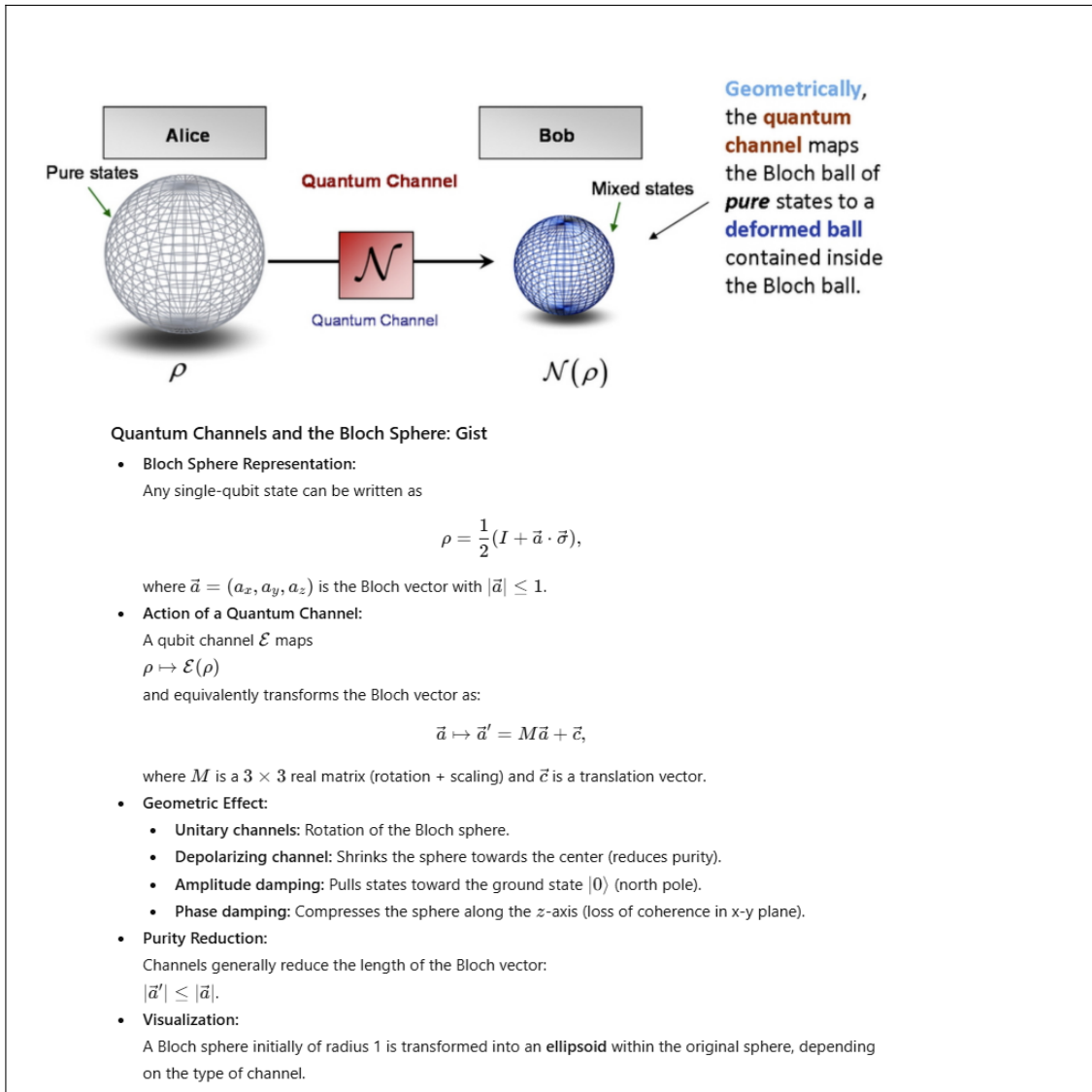


Figure 1: Effect of a quantum channel on the Bloch sphere. Pure states on the surface are mapped to mixed states inside a deformed Bloch ball.

4.1 Before the Channel (Alice)

- Alice prepares a pure state ρ .
- Geometrically, this corresponds to a point on the surface of the Bloch sphere.
- All pure states form a perfect unit sphere.

4.2 After the Channel (Bob)

After passing through the quantum channel:

- The Bloch sphere is deformed into an ellipsoid.
- Pure states generally become mixed.
- The output states lie inside a smaller region of the Bloch ball.

This deformation represents:

- Loss of coherence
- Noise and decoherence
- Irreversible interaction with the environment

5 Physical Meaning

- The shrinking of the sphere corresponds to loss of purity.
- The displacement of the center corresponds to biased noise.
- Only unitary channels preserve the full Bloch sphere.

Examples

- **Depolarizing channel:** shrinks the sphere uniformly.
- **Amplitude damping:** shifts and compresses the sphere.
- **Phase damping:** flattens the sphere along one axis.

6 Summary

- The Bloch sphere provides a geometric picture of qubit states.
- Quantum channels act as affine transformations.
- Pure states become mixed due to noise.
- The image of the Bloch sphere is always a smaller, deformed ball.